

MCQs on RESTful Web Services with Spring Boot

1. What does REST stand for?

- A) Representational State Transfer
- B) Real-time State Transfer
- C) Resource State Transfer
- D) Remote State Transfer

Answer: A) Representational State Transfer

2. Which HTTP methods are commonly used in RESTful services?

- A) GET, POST, PUT, DELETE
- B) GET, POST, PATCH, OPTIONS
- C) GET, POST, PUT, DELETE, PATCH
- D) All of the above

Answer: D) All of the above

3. What is a key principle of REST?

- A) Stateful communication
- B) Resource-based architecture
- C) Use of proprietary protocols
- D) None of the above

Answer: B) Resource-based architecture

4. Which of the following is NOT an advantage of using Spring Boot?

- A) Auto-Configuration
- B) Embedded Server
- C) Requires extensive manual configuration
- D) Rapid Development

Answer: C) Requires extensive manual configuration

5. What is the purpose of the `@RestController` annotation in Spring Boot?

- A) To define a service layer
- B) To create a RESTful web service controller
- C) To configure application properties
- D) To manage database connections

Answer: B) To create a RESTful web service controller

6. In Spring Boot, which dependency is essential for building web applications?

- A) Spring Data
- B) Spring Web
- C) Spring Security
- D) Spring Batch

Answer: B) Spring Web

7. What is the default port for a Spring Boot application?

- A) 8080
- B) 8000
- C) 3000

- D) 5000

Answer: A) 8080

8. Which of the following is a valid way to create a new user in the UserController?

- A) Using a GET request
- B) Using a POST request
- C) Using a DELETE request
- D) Using a PUT request

Answer: B) Using a POST request

9. What does the @GetMapping annotation do in Spring Boot?

- A) It maps HTTP GET requests to a specific handler method
- B) It maps HTTP POST requests to a specific handler method
- C) It maps HTTP DELETE requests to a specific handler method
- D) It maps HTTP PUT requests to a specific handler method

Answer: A) It maps HTTP GET requests to a specific handler method

10. Which of the following is a characteristic of SOAP?

- A) Uses JSON exclusively
- B) Stateless communication
- C) Built-in security features
- D) Primarily uses HTTP

Answer: C) Built-in security features

11. What is the main difference between SOAP and REST?

- A) SOAP is faster than REST
- B) REST is protocol-based, while SOAP is an architectural style
- C) REST uses multiple formats, while SOAP uses XML only
- D) SOAP is more flexible than REST

Answer: C) REST uses multiple formats, while SOAP uses XML only

12. Which of the following is a valid response status code for a successful GET request?

- A) 404
- B) 500
- C) 200
- D) 403

Answer: C) 200

13. What is the purpose of the @RequestBody annotation in Spring Boot?

- A) To bind request parameters to a method argument
- B) To specify the response body
- C) To bind the HTTP request body to a Java object
- D) To define the request URL

Answer: C) To bind the HTTP request body to a Java object

14. Which tool is commonly used to test RESTful APIs?

- A) Eclipse
- B) IntelliJ IDEA
- C) POSTMAN

- D) NetBeans

Answer: C) POSTMAN

15. What is the role of the `application.properties` file in a Spring Boot application?

- A) To define the database schema
- B) To configure application settings
- C) To manage user authentication
- D) To handle error messages

Answer: B) To configure application settings

16. Which of the following is a valid way to delete a user in the `UserController`?

- A) Using a GET request
- B) Using a POST request
- C) Using a DELETE request
- D) Using a PUT request

Answer: C) Using a DELETE request

17. What does the `@PutMapping` annotation do in Spring Boot?

- A) It maps HTTP PUT requests to a specific handler method
- B) It maps HTTP POST requests to a specific handler method
- C) It maps HTTP DELETE requests to a specific handler method
- D) It maps HTTP GET requests to a specific handler method

Answer: A) It maps HTTP PUT requests to a specific handler method

18. Which of the following is a common format used for data exchange in RESTful services?

- A) XML
- B) JSON
- C) HTML
- D) All of the above

Answer: D) All of the above

19. What is the purpose of the `@PathVariable` annotation in Spring Boot?

- A) To bind request parameters to a method argument
- B) To extract values from the URI
- C) To specify the response body
- D) To define the request URL

Answer: B) To extract values from the URI

20. Which of the following statements is true about RESTful services?

- A) They are stateful.
- B) They can only return XML.
- C) They are resource-based.
- D) They require a specific protocol.

Answer: C) They are resource-based.